Relativity Check

Seeing Things - or Moving Them?

States, transitions and the observer behind the measurement

Joern Christian Anders
B.Sc. Process Technology

6 November 2026
Working paper 1.0.0; not peer reviewed

Abstract

A changing detector record may reflect physical evolution, observer motion, propagation, occlusion, instrumental response or their combination. This paper develops an observer-aware framework motivated by the UNIQUANT state and phase representation. It separates a retained physical state, a dynamical law and an observation operator. A smooth Gaussian counterexample shows that identical initial position and energy distributions need not imply identical subsequent evolution. The observer's worldline, local frame, clock and detector response are then included explicitly. Local relativistic energy is distinguished from a passive coordinate change, and a quantum measurement model distinguishes outcome probabilities from measurement backaction. Two inverse-problem examples identify limitations of relative-position records and of transmission rays directed through one fixed point. Four reproducible analytic and synthetic diagnostics illustrate the results. A controlled measurement programme is proposed to distinguish target motion, observer motion and additional physical coupling. A conditional finite-occupancy extension separates capacity, composition, excitation and invariant mass. The framework does not establish that observed motion in general is an illusion, nor that position and scalar energy alone describe every physical process.

The question. When an image moves, what actually changed? The target, the observer, the path between them, the detector, or more than one of these?

The proposed organization.

$$\text{retained state } X \quad \xrightarrow{\text{dynamics}} \quad X(t), \qquad y_O(\tau) = \mathcal{M}[X; \mathcal{O}](\tau) + \eta_O. \tag{1}$$

Here $\mathcal{O}$ describes the observer and instrument; $\mathcal{M}$ includes propagation and measurement; η_O represents an explicitly modeled residual. This is an organizing statement, not a complete physical law until these objects have been specified.

Attribution and scope. UNIQUANT supplies a particular time/phase representation [1]; the earlier working papers motivate the distinction between dynamics and observation [2, 3]. Master equations, quantum measurement theory, relativistic observer frames and attenuation are established ingredients. Their use here is not a priority claim for those theories.

1 Position and energy do not generally identify a state

An economical state description must retain enough information to predict every observable claimed within its declared model. Finer resolution of retained variables cannot reconstruct information that the representation omits.

For a nonrelativistic free particle in one spatial dimension, let $H = \hat{p}^2/(2m)$ and $\sigma, k > 0$. Define two initial states

$$\psi_\pm(x,0) = \frac{1}{(2\pi\sigma^2)^{1/4}} \exp\left[-\frac{x^2}{4\sigma^2}\right] e^{\pm ikx}. \tag{2}$$

Both are smooth, normalized and have finite moments of all orders.

Proposition 1.1 (Equal initial marginals, different evolution)**.** *The states* (2) *have identical initial position distributions and identical complete energy distributions, but opposite initial probability currents and different subsequent position distributions.*

Proof. Their common position density is

$$n_0(x) = |\psi_\pm(x,0)|^2 = \frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}. \tag{3}$$

Their momentum distributions are Gaussians centered at $\pm\hbar k$ with common variance $\hbar^2/(4\sigma^2)$. Reflection $p \mapsto -p$ exchanges these distributions and leaves $p^2/(2m)$ unchanged. Thus their energy distributions agree, including

$$\langle E\rangle_\pm = \frac{\hbar^2 k^2}{2m} + \frac{\hbar^2}{8m\sigma^2}. \tag{4}$$

The standard free-particle current [4] gives

$$j_\pm(x,0) = \frac{\hbar}{m}\,\mathrm{Im}(\psi_\pm^* \partial_x \psi_\pm) = \pm\frac{\hbar k}{m} n_0(x). \tag{5}$$

Free evolution has centers $\langle x\rangle_\pm = \pm\hbar k t/m$ and densities

$$n_\pm(x,t) = \frac{1}{\sqrt{2\pi}\sigma_t} \exp\left[-\frac{(x \mp \hbar k t/m)^2}{2\sigma_t^2}\right], \qquad \sigma_t^2 = \sigma^2 + \left(\frac{\hbar t}{2m\sigma}\right)^2. \tag{6}$$

For $t > 0$ the two centers differ, so the position distributions differ. □

The missing information is directed momentum or, equivalently in this example, the spatial phase gradient. Fixing the same observer for both preparations does not remove this counterexample. A proposed transition law can carry the missing information, but then the law contains more than position and scalar energy data.

These are *separately defined* initial measurement distributions. No ordinary positive simultaneous distribution $P(x,E)$ is assumed: position and free-particle energy generally do not commute. The later densities refer to identically prepared ensembles, not to a particle after a destructive initial position measurement. This argument tests descriptive sufficiency; it does not decide every interpretation of whether unobserved positions exist.

2 The observer is a worldline, frame, clock and instrument

Use coordinates $x^\mu = (ct, \mathbf{x})$, metric signature $(-+++)$, and an observer worldline $z_O^\mu(\tau)$ parameterized by proper time. Its four-velocity is

$$u_O^\mu = \frac{\mathrm{d}z_O^\mu}{\mathrm{d}\tau}, \qquad g_{\mu\nu} u_O^\mu u_O^\nu = -c^2. \tag{7}$$

The worldline determines the velocity. Listing both as independently fitted inputs would require a consistency constraint. A local orthonormal tetrad satisfies

$$e_{(0)}^\mu = u_O^\mu / c, \qquad g_{\mu\nu} e_{(a)}^\mu e_{(b)}^\nu = \eta_{ab}. \tag{8}$$

Its spatial axes encode the observer's physical orientation. Clock calibration, spectral response, aperture, angular response and exposure complete the experimental description. A single fixed coordinate point is not this full description.

2.1 Local energy and momentum

For a four-momentum $p^\mu = (E/c, \mathbf{p})$ at the observer's event,

$$\boxed{E_O = -p_\mu u_O^\mu}, \qquad p_{O(i)} = p_\mu e_{(i)}^\mu. \tag{9}$$

In a common inertial coordinate system this becomes

$$E_O = \gamma_O (E - \mathbf{v}_O \cdot \mathbf{p}), \qquad \gamma_O = (1 - v_O^2/c^2)^{-1/2}. \tag{10}$$

The contraction in (9) is coordinate invariant for the specified particle and observer. Changing the physical observer velocity can change E_O; merely relabeling coordinates while transforming both vectors cannot. This distinction is part of established relativity [5].

For a one-dimensional boost,

$$E' = \gamma(E - vp), \qquad p' = \gamma(p - vE/c^2), \qquad E'^2 - c^2 p'^2 = E^2 - c^2 p^2 = m^2 c^4. \tag{11}$$

An energy value alone therefore does not retain the direction needed to predict what a moving observer measures.

2.2 Remote emission is not a local measurement

For a detected photon,

$$h\nu_O = -p_{\mu,\mathrm{det}} u_O^\mu. \tag{12}$$

The momentum must first be propagated to the detection event. In vacuum geometric optics on a specified spacetime, this uses the appropriate null propagation law. In a medium, its optical response must also be modeled. Four-vectors at remote events belong to different tangent spaces; subtracting them in curved spacetime requires a specified identification or transport construction.

Observer dependence is therefore not arbitrary visual distortion. It is a calculable relation between a physical state, propagation and a local measurement. Selecting a common time origin also does not establish identical clock rates or a global observer-independent present.

3 Dynamics, propagation and detection are different maps

For a classical or coarse-grained state, (1) may be made concrete as a composition of evolution, propagation, exposure and detector response. A position in a reconstructed image is an output of this chain, not automatically a direct three-dimensional coordinate of the emitter.

3.1 A causal camera model

In a flat-spacetime vacuum model, an emission event on a target trajectory and a detection event on the camera trajectory satisfy

$$\mathbf{q} = \mathbf{x}(t_{\mathrm{em}}) - \mathbf{b}(t_{\mathrm{det}}), \qquad c(t_{\mathrm{det}} - t_{\mathrm{em}}) = |\mathbf{q}|. \tag{13}$$

The detected propagation direction is opposite to the source-bearing vector $\mathbf{q}/|\mathbf{q}|$. Transform the local photon momentum to the camera frame before applying its orientation and projection. At appreciable relative speed, rotation alone does not replace the Lorentz transformation and aberration.

At low speeds, small angles and known approximately constant depth r, a useful one-axis approximation is

$$\theta(t) = \frac{x_\perp(t) - b_\perp(t)}{r} - \phi(t) + \eta_\theta(t). \tag{14}$$

The angle ϕ describes camera pointing, with the displayed sign convention. Equation (14) assumes that synchronization and light-time handling permit the stated common-time approximation. It is not a cosmological distance law.

3.2 Quantum outcome probabilities and backaction

For a specified quantum state ρ and observer/instrument arrangement,

$$\Pr(a \mid \mathcal{O}) = \mathrm{Tr}[\rho E_a^{(\mathcal{O})}], \qquad E_a^{(\mathcal{O})} \geq 0, \quad \sum_a E_a^{(\mathcal{O})} = I. \tag{15}$$

Propagation can be absorbed into these effects, or represented explicitly by a quantum channel before detection. Loss must include a no-detection outcome or an explicit conditioning rule. This is standard quantum measurement language [6].

A POVM gives outcome probabilities. Predicting backaction additionally needs an instrument, for example

$$\mathcal{I}_a(\rho) = \sum_r M_{ar}\rho M_{ar}^\dagger, \quad E_a = \sum_r M_{ar}^\dagger M_{ar}, \quad \rho_a' = \frac{\mathcal{I}_a(\rho)}{\Pr(a)}. \tag{16}$$

No universal relocation of every particle at each recorded observation is implied. The physical interaction and the conditional update of a description must be specified separately.

Under a consistent unitary change of representation, $\rho' = U\rho U^\dagger$ and $E_a' = UE_aU^\dagger$, cyclicity of the trace gives $\mathrm{Tr}(\rho' E_a') = \mathrm{Tr}(\rho E_a)$. This algebraic representation check is not a claim that arbitrary spacetime transformations are implemented by a finite matrix.

4 When target motion and observer motion look alike

Proposition 4.1 (Ambiguity of relative-position data)**.** *In the synchronized Newtonian measurement model $y(t) = x(t) - b(t)$, the measured function y does not uniquely determine both unknown trajectories x and b.*

Proof. For any common function $h(t)$, the pair

$$x'(t) = x(t) + h(t), \qquad b'(t) = b(t) + h(t) \tag{17}$$

produces $x' - b' = x - b$. □

This is an inverse-problem ambiguity for that measurement model. An arbitrary accelerated h is not a symmetry of fixed equations of motion with unchanged forces. Independent pose information, dynamical constraints or additional measurements can reduce the ambiguity.

4.1 Three different distance scalings

For a stationary target and transverse observer oscillation $b_\perp = A \sin \omega t$, (14) gives, with fixed pointing,

$$\theta(t) \simeq -\frac{A}{r} \sin \omega t. \tag{18}$$

A fixed physical translation produces a *smaller angular amplitude* at larger distance. A fixed pointing-angle uncertainty instead gives

$$\sigma_\perp \simeq r\sigma_\theta. \tag{19}$$

For small-angle parallax $\varpi \simeq B/r$, fixed baseline B and sufficiently small angular uncertainty, local propagation gives

$$\sigma_r \simeq \frac{r^2}{B}\sigma_\varpi. \tag{20}$$

These concern different mechanisms and different measured quantities. Neither establishes a universal distance beyond which all positions or motions are false. The reciprocal-parallax approximation also requires adequate disparity signal and an appropriate inference model [3].

4.2 A calibrated reconstruction

With independently measured translation and pointing, define

$$\widehat{x} = r(\theta + \phi_{\text{cal}}) + b_{\text{cal}}. \tag{21}$$

For exact r, unbiased independent errors and the correct linear model,

$$\sigma_x^2 = r^2(\sigma_\theta^2 + \sigma_\phi^2) + \sigma_b^2. \tag{22}$$

This follows because the error is $r\eta_\theta + r(\phi_{\text{cal}} - \phi) + (b_{\text{cal}} - b)$. Unknown range, timing errors and correlated calibration add terms and may change identifiability. Two cameras do not automatically remove unknown common motion; their poses and baseline must themselves be constrained.

5 Many directions do not automatically reveal every depth

The presence of particles is not a universal hard stop for light. For a narrow, monoenergetic direct beam with scattering out of the beam treated as attenuation, the ideal transmission model is [7]

$$\frac{I(L,\nu)}{I_0(\nu)} = \exp\left[-\int_L \mu(\nu,\mathbf{x})\,\mathrm{d}s\right]. \tag{23}$$

The coefficient μ has units of inverse length and depends on material and photon energy. A broad detector that also accepts scattered radiation needs a more complete transport model. Very small transmission can create a practical detection limit for a given source, exposure and detector; it is not a universal geometric edge of space.

Proposition 5.1 (Central-ray nonuniqueness)**.** *Even all transmission measurements along straight lines through a single fixed point fail to determine a general three-dimensional attenuation field uniquely.*

Proof. Take a sphere of radius R centered at that point. With $\mu_0 > 0$, define

$$\mu_1(r) = \mu_0 \quad (0 \le r < R), \qquad \mu_2(r) = \begin{cases} 1.5\mu_0, & 0 \le r < R/2, \\ 0.5\mu_0, & R/2 \le r < R, \end{cases} \tag{24}$$

and set both to zero outside the sphere. Every central diameter has optical depth

$$\tau_1 = 2\mu_0 R, \qquad \tau_2 = 1.5\mu_0 R + 0.5\mu_0 R = 2\mu_0 R. \tag{25}$$

The fields differ but yield identical central transmissions. Every half-ray from the surface to the center also has the same integral in both models. □

For impact parameter $0 \le b < R$, the off-axis optical depths are

$$\tau_1(b) = 2\mu_0\sqrt{R^2 - b^2}, \tag{26}$$

$$\tau_2(b) = \mu_0\sqrt{R^2 - b^2} + 2\mu_0\sqrt{\max(R^2/4 - b^2, 0)}. \tag{27}$$

Thus appropriately offset rays can distinguish these examples. If $\mu(\nu,\mathbf{x}) = \kappa(\nu)\varrho(\mathbf{x})$ for the same material, equal central density columns remain indistinguishable at every frequency. Additional frequencies help when they supply genuinely independent material information.

For equal material composition, these two profiles have different total masses: the second has $5/8$ of the first. An independent total-mass measurement could therefore distinguish them. The proposition concerns the specified transmission data, not all possible observations. Multiple directions, frequencies and baselines must be selected for the actual measurement operator and the retained model class.

6 What UNIQUANT can organize, and what must be added

The current UNIQUANT construction defines the rational time unit and restored time address [1]

$$q = \frac{75}{1012}\,\mathrm{s}, \qquad t_n - t_{\mathrm{ref}} = q(A_n + \rho_n), \qquad -\tfrac{1}{2} \le \rho_n < \tfrac{1}{2}. \tag{28}$$

Keeping the remainder retains the represented model time. Dropping it leaves a bounded rounding error of at most $q/2$ per address. This is a representation result, not a measured fundamental waiting time for physical particle transitions.

The 35 unequal 13/23 classes are cycle-phase bins, not automatically spatial cells or quantum phases. For bin masses p_j, widths w_j and midpoints m_j, the piecewise-constant reconstruction has moments

$$\widetilde{c}_k = \sum_j p_j e^{2\pi i k m_j} \operatorname{sinc}(\pi k w_j), \qquad \operatorname{sinc} z = \frac{\sin z}{z}, \quad \operatorname{sinc} 0 = 1. \tag{29}$$

Using the bin index as an equally spaced angle would discard the actual geometry. Finite bin masses also do not by themselves retain every within-bin mode.

6.1 A local probabilistic extension

For a finite coarse model, let C include spatial occupations, directed momentum classes, internal energies and whatever field or wall state the chosen closure requires. A possible statistical evolution is

$$\partial_t P(C,t) = \sum_{C' \neq C} \left[w(C' \to C) P(C',t) - w(C \to C') P(C,t) \right]. \tag{30}$$

Finite nonnegative rates preserve normalization by cancellation of the paired sums. This is a model ansatz; geometry and class labels do not determine its rates. Required conservation laws must constrain each internal transition. For example,

$$\Delta N = 0, \qquad \Delta \mathbf{\Pi} = 0, \qquad \Delta E = 0 \tag{31}$$

for closed internal collisions. Wall, reservoir and external-force exchanges need their corresponding balances. A probability normalization is not energy in joules.

Local grid rules can reproduce hydrodynamic behavior under suitable conditions [8]; arbitrary conservative rules need not have the correct isotropy, equilibrium, viscosity or heat conductivity. One-particle marginals may also require correlation closures. A positive classical kernel alone does not retain general quantum coherence; equations (15)–(16) require a quantum state and compatible dynamics when coherent effects are claimed.

For the heated-pipe application [2], energy enters through specified boundary interactions or source terms. Temperature requires an appropriate state relation, not simply an identification with every unresolved counterflow. Any new cosmic contribution would need an independent coupling law and a measured energy and momentum budget. It does not follow from the time address.

7 Four reproducible diagnostics

Figure 1 is generated by the accompanying `reproduce.py`. Every panel uses analytic laws or declared synthetic inputs. None is a fit to astronomical, pipe-flow or quantum-optics observations.

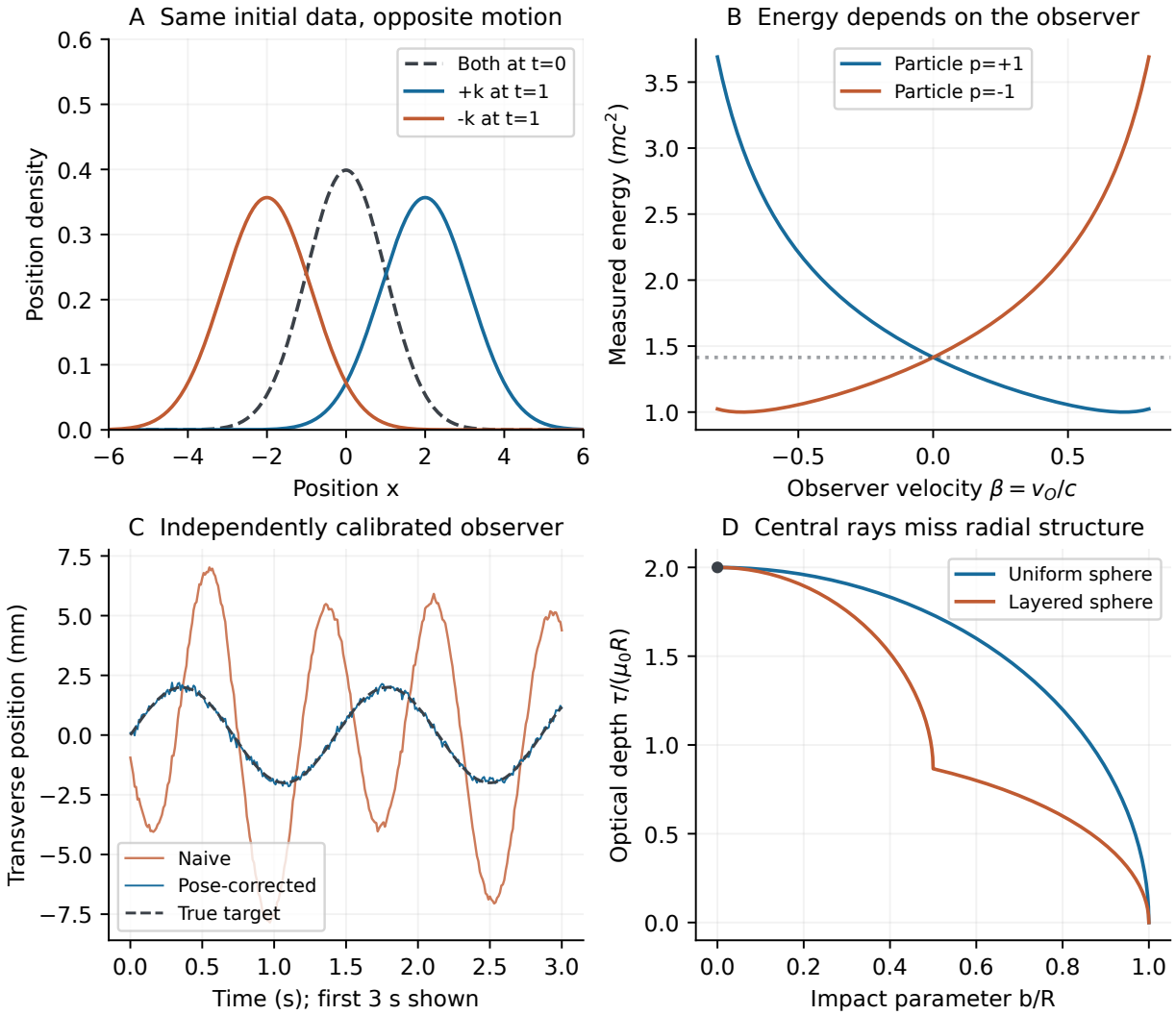


Figure 1: Analytic and synthetic illustrations. **A:** The common initial position density and the opposite later motions of Section 1. **B:** Local relativistic energies for opposite particle momenta and a variable observer. **C:** A simulated camera record before and after correction using independent pose calibrations; only the first three seconds are drawn. **D:** Central transmission agrees for distinct radial profiles; offset transmission separates them. Numerical details and units are specified in the text and source bundle.

Panel A. Set $\hbar = m = 1$, $\sigma = 1$ and $k = 2$. Both initial energies are $17/8$, while the centers at $t = 1$ are $+2$ and -2, with common variance $5/4$.

Panel B. Set $m = c = 1$, $p = \pm 1$ and $E = \sqrt{2}$. The curves are $E_\pm(\beta) = (\sqrt{2} \mp \beta)/\sqrt{1-\beta^2}$. The programme checks the mass-shell identity for both branches, rather than only drawing transformed energies.

Panels C and D. The camera diagnostic uses the exact stated paraxial model with known range and independent Gaussian calibration errors. It tests that specific correction, not the recoverability of arbitrary unknown motion. The sphere diagnostic evaluates (26) directly, with $R = \mu_0 = 1$.

8 Numerical results and their interpretation

The deterministic diagnostics use fixed parameters and seeds, with numerical quadrature only as a check on the Gaussian energy moment. The results below are generated from the same calculation as Figure 1.

Diagnostic	Result
A: equal mean energies ($\hbar = m = 1$)	$17/8 = 2.125$
A: centres at $t = 1$	$+2,\ -2$
B: E_O at $\beta_O = 0.6$, $p = +1$	1.01776695
B: E_O at $\beta_O = 0.6$, $p = -1$	2.51776695
B: maximum mass-shell residual	7.1×10^{-15}
C: naive position RMSE	3.6074 mm
C: corrected position RMSE	0.1044 mm
C: predicted standard uncertainty	0.1039 mm
C: fraction inside $\pm 1.96\sigma_x$	0.9550
D: central optical depth, both profiles	2
D: off-axis depths at $b/R = 0.75$	1.322876, 0.661438

Table 1: Synthetic or analytic diagnostics. Numerical residuals measure the implementation of these finite examples, not empirical accuracy of a new theory.

8.1 Camera parameters and uncertainty

There are 1001 samples on $0 \le t \le 10\,\mathrm{s}$, with known exact range $r = 10\,\mathrm{m}$. Inputs are

$$x(t) = 2\,\mathrm{mm}\sin(2\pi\,0.7\,\mathrm{Hz}\,t), \tag{32}$$

$$b(t) = 5\,\mathrm{mm}\sin(2\pi\,1.3\,\mathrm{Hz}\,t), \tag{33}$$

$$\phi(t) = 100\,\mu\mathrm{rad}\cos(2\pi\,1.1\,\mathrm{Hz}\,t). \tag{34}$$

Independent Gaussian angle, translation-calibration and pointing-calibration errors have standard deviations $10\,\mu$rad, $20\,\mu$m and $2\,\mu$rad, with seeds 23, 24 and 25, respectively. The target and observer motions are intentionally different. The corrected record retains the target motion while reducing the specified observer artifacts.

The predicted standard uncertainty is from (22). The realized RMSE and interval coverage fluctuate with the finite noise sample. They are not guaranteed to equal their expectation exactly. Pose errors are independent inputs here, rather than fitted from the same target image. Known range and the correct forward model are deliberate assumptions, not results of this experiment.

Panel D's central equality remains exact under frequency-factorized attenuation, but its off-axis inequality and different total masses provide examples of useful additional measurements. This illustrates why an identifiability claim must always name the available measurement family.

9 A test programme that can distinguish the explanations

Before fitting data, specify the physical states, their dynamics, the observation operator, independently calibrated quantities, noise correlations and the claimed validity interval. A freely adjustable observer trajectory can absorb real target motion just as a freely adjustable target trajectory can absorb observer errors.

Question	Controlled change	Discriminating record
Target or camera motion?	Move one calibrated stage at a time; record camera pose and target position independently.	Raw angles, exposure times, stage encoders and pose uncertainties.
Translation or pointing?	Change known target range and imposed camera rotation separately.	Translational angle scales as $1/r$; imposed pointing angle does not.
Projection or depth?	Change calibrated baseline and include independent range data where available.	Agreement of raw-data predictions across geometries, not only a reconstructed image.
Occlusion or state change?	Use offset paths, several bands and measured source intensity.	Transmission, scattering or emission described by a fixed material/transport model.
Missing phase information?	Prepare the two states of Section 1 in a suitable quantum implementation.	Independent ensemble position, momentum/energy and later-position statistics.
Additional energy coupling?	Change field, material or orientation under thermal and mechanical controls.	Predicted response with an accounted energy and momentum exchange.

Table 2: Proposed tests, not experiments performed in this paper. Quantum and relativistic examples need suitable physical implementations; an ordinary camera bench does not test every part of the framework.

9.1 Comparison on observations that were not used for fitting

Let M_0 include calibrated observer motion, propagation and established material response. Let M_1 add one explicitly specified coupling with fixed units and parameters. Estimate parameters on a declared training/calibration interval; then predict later raw records, new viewing geometries or controlled orientations. For a correctly declared Gaussian observation model, one possible score is

$$-2\log L_M = (y-\mu_M)^T\Sigma_M^{-1}(y-\mu_M) + \log\det\Sigma_M + \text{constant}. \tag{35}$$

Other noise models require their own likelihoods. Compare held-out predictive performance and uncertainty coverage, with parameter freedom and calibration uncertainty accounted for. Retuning phases or observer trajectories after seeing each test record does not establish a new physical explanation.

10 What has been established, and what remains open

The title asks a deliberately pointed question. The answer depends on the measurement: recording a changing image is not the same operation as exerting a force on its apparent source. An actual interaction may change the source, but that requires a specified coupling and an energy/momentum balance.

Established within the stated models	Not established by these results
Smooth states can share initial position and energy marginals yet evolve differently.	Every process can be reduced to scalar energy and location alone.
Local measured energy depends on the physical observer.	A passive coordinate relabeling supplies or removes physical energy.
Unknown target and observer trajectories can be ambiguous from relative-position data alone.	All observed motion is caused by an unrecognized observer oscillation.
Central-ray transmission has an explicit spatial null ambiguity.	No observer can ever obtain useful information at large distance.
Calibrated pose correction works in the declared synthetic model.	Existing astronomical distance estimates or physical laws have been refuted.
UNIQUANT can organize retained time and phase information.	Its address unit is a universal physical jump time or its bins are a complete quantum state.

Table 3: Logical scope of the derivations and synthetic examples.

10.1 The constructive research claim

An observer-aware state-transition framework can organize physical evolution and measurement in one description. Its sufficiency depends on retained state variables, interaction laws and measurement operators. Direction, phase and correlations can be encoded explicitly or inside a conditional dynamical law; moving them into that law does not eliminate their information content.

The next concrete step is a restricted forward model with independently fixed transition or interaction parameters. For example, a heated-pipe model could predict outlet temperature, wall heat transfer and pressure drop while a separate measurement model predicts the sensors' outputs. A camera bench could instead test recovery of known target motion under independently recorded observer motion. These are different applications with different closures and validation data.

Interpretation. A single local detection is compatible with a local event record. It does not, by itself, establish the complete state, a unique unobserved trajectory or an absence of transport. Conversely, incomplete observation should be represented by explicit uncertainty and nonuniqueness, rather than silently filled with a preferred physical story.

A A finite-occupancy hypothesis: what would “100 percent” mean?

The additional proposal is that a spatial region may have a maximum number of constituents, with its observable properties determined by occupancy, charge, orientation, grouping and excitation. A finite-capacity model can express part of this idea. The existence of a universal cosmic capacity is an additional hypothesis, not a consequence of probability normalization.

Choose finite cell capacities $K_i > 0$ and restrict the configuration space to

$$N_i = \sum_\alpha N_{i\alpha} \le K_i. \tag{36}$$

The state class α may include species, charge, orientation, momentum and internal excitation. Define capacity fractions, filling and vacancy by

$$c_{i\alpha} = \frac{\mathbb{E}[N_{i\alpha}]}{K_i}, \qquad f_i = \sum_\alpha c_{i\alpha}, \qquad v_i = 1 - f_i, \qquad v_i + \sum_\alpha c_{i\alpha} = 1. \tag{37}$$

This produces a precise 100 percent bookkeeping convention. Charge, orientation and excitation are attributes of the occupied classes, not additional fractions that can be added once more to the filling fraction. Composition among the present particles is a different normalization:

$$x_{i\alpha} = \frac{\mathbb{E}[N_{i\alpha}]}{\mathbb{E}[N_i]}, \qquad \sum_\alpha x_{i\alpha} = 1 \quad \text{if } \mathbb{E}[N_i] > 0. \tag{38}$$

To leading nonrelativistic order with specified species masses,

$$\varrho_i \simeq \frac{1}{V_i} \sum_\alpha m_\alpha \mathbb{E}[N_{i\alpha}]. \tag{39}$$

Equal filling need not mean equal mass density if masses or physical cell volumes differ. Ordinary molecular species also differ in nuclear composition. Changing orientation or electron occupation does not turn one proton number into another; common elementary constituents do not make all molecules identical [9].

A.1 A physical capacity must survive a change of representation

A fixed K assigned to every freely resized numerical cell creates a grid-dependent density ceiling. A proposed universal proper-number-density bound would instead need a dimensional parameter such as $\kappa_{\max}$ in m^{-3}, a precise definition of constituent, and consistent scaling $K_i \simeq \kappa_{\max} V_i$ over the declared coarse-graining regime. Reactions and changes of constituent species must specify what is being counted.

An observer-dependent density is not automatically an invariant bound. For a single conserved timelike number current $\mathcal{N}^\mu = n_* U^\mu$, the locally measured number density is

$$n_O = -\frac{\mathcal{N}^\mu u_{O\mu}}{c^2}, \qquad n_* = \frac{\sqrt{-\mathcal{N}_\mu \mathcal{N}^\mu}}{c}. \tag{40}$$

The rest density n_* and moving-observer density n_O can differ. This formula assumes the stated conserved fluid current; it is not a universal definition of particle number for all quantum fields. A physical capacity hypothesis must name its invariant or preferred-frame content and make predictions beyond a chosen bin size.

A.2 Ground state, excitation, density and mass are separate quantities

For a Hamiltonian with a specified ground energy E_0, one may define

$$E_{\mathrm{exc}} = E - E_0 \geq 0. \tag{41}$$

This sets an excitation reference. It does not remove the rest energy, binding energy or zero-point contributions of the material. For example, a harmonic phonon model has [10]

$$H_{\mathrm{ph}} = \sum_{\lambda} \hbar\omega_{\lambda}(\hat{n}_{\lambda} + \tfrac{1}{2}). \tag{42}$$

Its ground state has no excited phonons but can have quantum zero-point motion. An ordered crystal is therefore not generally an object with zero total energy or perfectly fixed microscopic constituent positions.

For the complete system,

$$M^2c^4 = E_{\mathrm{tot}}^2 - c^2|\mathbf{P}_{\mathrm{tot}}|^2, \qquad E_{\mathrm{rest}} = Mc^2. \tag{43}$$

Comparing the same closed material system in its center-of-momentum frame,

$$\Delta M = \frac{\Delta E_{\mathrm{rest}}}{c^2}. \tag{44}$$

Positive net stored energy increases its invariant mass. Energy exported during binding can reduce the mass left in a sample. Its bulk density can decrease while its mass increases if the volume expands sufficiently:

$$\frac{\Delta\varrho}{\varrho} \simeq \frac{\Delta M}{M} - \frac{\Delta V}{V} \qquad \text{for small changes of a uniform body.} \tag{45}$$

Consequently no universal rule "more excitation means less mass" follows from filling, temperature or probability normalization.

A.3 Entropy bounds and proposed fundamental frequencies

A different proposal in gravitational physics is the Bekenstein bound [11], conventionally written for suitable bounded systems as

$$S \leq \frac{2\pi k_B ER}{\hbar c}, \qquad \frac{S}{k_B \ln 2} \leq \frac{2\pi ER}{\hbar c \ln 2}. \tag{46}$$

This concerns entropy and system energy/size under its applicable assumptions, not a direct count of particles or an unconditional volume-packing constant. Quantum-field formulations require carefully specified entropy and energy differences; relative entropy provides a precise version in appropriate settings [12]. A freely zeroed excitation energy must not silently replace the energy required by a particular bound.

Candidate excitation frequencies could enter an explicit driving Hamiltonian, $H_{\mathrm{drive}}(t) = \sum_{\ell} A_{\ell}G_{\ell}\cos(\omega_{\ell}t + \phi_{\ell})$, with $A_{\ell}G_{\ell}$ in energy units and a defined energy-supplying source. Charge and orientation may affect G_{ℓ}. Neither capacity normalization nor a cycle label selects a unique fundamental frequency.

Testable extension. Fix the constituent definition, capacities, interactions and couplings before comparison. Require stability under grid refinement and predictions of composition, density, heat uptake and mass/energy balance across materials and orientations. A finite-capacity parametrization is available; its proposed cosmic universality and any mass-decreasing excitation law remain separate empirical claims.

A.4 Binding, constraints and evaporation

Intermolecular interactions, energetic barriers and external boundaries can stabilize structures. A universal rigid grid is not required: an ordinary liquid does not possess a permanent crystal lattice, yet its molecules interact and its surface has an energetic cost. Evaporation transfers molecules out of the liquid; it need not change their chemical identity or wait for the boiling temperature.

A possible coarse description uses number density $n(\mathbf{x})$ and a structural order parameter $\eta(\mathbf{x})$ at fixed temperature. The latter labels a declared degree of order; it is not the quantum phase or a complete crystal structure. Following the established gradient-free-energy approach [13], propose

$$F[n,\eta;T] = \int_\Omega \left[f_{\mathrm{bulk}}(n,\eta,T) + \frac{\kappa_n}{2}|\nabla n|^2 + \frac{\kappa_\eta}{2}|\nabla \eta|^2 \right] \mathrm{d}V. \tag{47}$$

Here f_{bulk} is free energy per volume and the positive gradient coefficients carry the units needed for F to be in joules. Bulk minima and gradient penalties describe phase preference and diffuse interfaces. Anisotropy, elasticity and actual lattice periodicity require further terms or fields.

For smooth F, a simple isothermal relaxation model is

$$\mu = \frac{\delta F}{\delta n}, \quad \chi = \frac{\delta F}{\delta \eta}, \qquad \partial_t n = \nabla \cdot (M \nabla \mu), \quad \partial_t \eta = -L\chi, \qquad M, L \geq 0. \tag{48}$$

With periodic or compatible no-flux boundary conditions, integration by parts gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega n \,\mathrm{d}V = 0, \qquad \frac{\mathrm{d}F}{\mathrm{d}t} = -\int_\Omega \left[M|\nabla \mu|^2 + L\chi^2 \right] \mathrm{d}V \leq 0. \tag{49}$$

Local populations and phase boundaries can therefore change while the total constituent count is conserved. The lost free energy is not deleted physical energy: the isothermal environment and its heat exchange belong in a full energy balance.

This relaxation example omits fluid advection and the nonisothermal energy equation; it is not a complete evaporation simulator. At planar bulk liquid-vapor coexistence in equilibrium, temperature, pressure and chemical potential match between phases. Predicting actual phase densities, vapor pressure and latent heat requires a material free energy and transport laws; for water, IAPWS-95 is an established thermodynamic reference [14].

A capacity constraint from Appendix A can be imposed on admissible densities or configurations. Its imposition does not derive the capacity from molecular physics. The constructive question is whether independently specified interactions and constraints predict phase behavior and energy exchange. The observation of evaporation alone establishes neither a universal grid boundary nor a cosmic fundamental frequency.

Reproduction and references

The source bundle contains `paper.tex`, `reproduce.py`, `results.json`, `camera_synthetic.csv`, the vector figure, `README.md` and `build.sh`. Run `python3 reproduce.py`, then compile `paper.tex` twice with pdfLaTeX. Required Python packages are NumPy and Matplotlib. Fixed seeds make the declared random inputs reproducible; tiny floating-point or rendering differences may occur across dependency versions. The JSON records parameters and checks. No observational data are supplied or implied. The complete source files, rather than an image of the figure, are the reproducible calculation.

References

[1] J. C. Anders, *Uniquant: rationale Zeit- und Phasenkoordinaten. Beschraenkte Quantisierungsfehler, geometrische Momente und identifizierbare Verteilungsrekonstruktion*. Working paper 1.0.0, 6 November 2026; German-language supplied manuscript, not peer reviewed.

[2] J. C. Anders, *all at once - maybe?* Working paper 1.0.0, 6 November 2026; supplied manuscript, not peer reviewed. Sections 5–7 discuss energy balance, material response and the observer.

[3] J. C. Anders, *Is Acceleration Quadratic - or Is It Depth Uncertainty?* Working paper 1.0.0, 6 November 2026; supplied manuscript, not peer reviewed. Used as related background, not as independent experimental evidence.

[4] Cornell University, *Applications of Quantum Mechanics: Wave Packets*. Course notes; amplitude, phase and probability current. https://quantum.lassp.cornell.edu/lecture/wave_packets.

[5] L. Amendola, *General Relativity*, version 2.9. Heidelberg University lecture notes, Chapter 1, Sections 1.11–1.12. https://www.thphys.uni-heidelberg.de/~amendola/teaching/gr2021.pdf.

[6] M. Kliesch and I. Roth, "Theory of Quantum System Certification," *PRX Quantum* **2**, 010201 (2021). https://doi.org/10.1103/PRXQuantum.2.010201.

[7] J. H. Hubbell and S. M. Seltzer, *Tables of X-Ray Mass Attenuation Coefficients and Mass Energy-Absorption Coefficients*, NIST Standard Reference Database 126, Section 2: The Mass Attenuation Coefficient. https://physics.nist.gov/PhysRefData/XrayMassCoef/chap2.html.

[8] U. Frisch, B. Hasslacher and Y. Pomeau, "Lattice-Gas Automata for the Navier-Stokes Equation," *Physical Review Letters* **56**, 1505–1508 (1986). https://doi.org/10.1103/PhysRevLett.56.1505.

[9] NIST, *Periodic Table of the Elements*, Special Publication 966; atomic properties and species information. https://www.nist.gov/pml/periodic-table-elements.

[10] D. Tong, *Statistical Physics*, Chapter 3: Quantum Gases, including phonons and zero-point energy. University of Cambridge lecture notes. https://www.damtp.cam.ac.uk/user/tong/statphys/statmechhtml/S3.html.

[11] J. D. Bekenstein, "Universal upper bound on the entropy-to-energy ratio for bounded systems," *Physical Review D* **23**, 287–298 (1981). https://doi.org/10.1103/PhysRevD.23.287.

[12] H. Casini, "Relative entropy and the Bekenstein bound," *Classical and Quantum Gravity* **25**, 205021 (2008). https://doi.org/10.1088/0264-9381/25/20/205021.

[13] J. W. Cahn and J. E. Hilliard, "Free Energy of a Nonuniform System. I. Interfacial Free Energy," *The Journal of Chemical Physics* **28**, 258–267 (1958). https://doi.org/10.1063/1.1744102.

[14] IAPWS, *Revised Release on the IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use*, R6-95(2018). https://iapws.org/technical-guidance/release/IAPWS-95.